



Control of unexpected wind-induced vibration of an in-service cable-stayed bridge using vortex generators

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ABSTRACT

Nine years after opening, a cable-stayed bridge with steel-concrete composite girder in China was found to experience unexpected large-amplitude wind-induced vibrations (WIV) under typical wind speeds (approximately 10 m/s). Aerodynamic configuration inspection and field measurement were conducted immediately to investigate the cause of it. Field measurement indicated that the structural damping of the bridge girder was as low as 0.2%–0.4%, differing significantly from the assumed values of the recognized design codes. The unexpected WIV, suspected to be vortex-induced vibration (VIV), was reproduced in wind tunnel tests on sectional models with scale ratios of 1:50 and 1:25, and the low structural damping was speculated to be the main cause of it. Also, aerodynamic solutions were further explored, as a result of which, vortex generators (VG) installed under the wind fairings were proposed, and the aerodynamic mechanism of them was investigated through computational fluid dynamics (CFD). Finally, VGs were installed on the actual bridge, and since then no further signs of unexpected WIV have been observed.

1. Introduction

Long-span bridges are characterized by structural flexibility and low damping, which makes them susceptible to various types of wind-induced vibrations (WIV) under wind loads. Among these WIVs, vortex-induced vibration (VIV) is the most common one (Fujino et al., 2019; Larsen and Larose, 2015). In recent years, numerous long-span bridges, in service or under construction, have experienced significant VIV (Battista and Pfeil, 2000; Corriols and Morgenthal, 2012; Fujino and Yoshida, 2002; Ge et al., 2022; Larsen et al., 2000). VIV is a self-limited oscillation caused by periodic vortex shedding around the structure. Although VIV does not lead to catastrophic failure like flutter, frequent and sustained VIV can induce fatigue damage to bridge components, affect the comfort of pedestrians and vehicles, and even trigger traffic safety incidents (Williamson and Raghuraman, 2004). During the construction phase, VIV poses risks to the safety of personnel and equipment, potentially delaying the construction schedule. Furthermore, significant VIV in bridges can lead to public panic and suspicion, with social repercussions that may surpass the actual impact on structural integrity and traffic safety

(Frandsen, 2001). For example, the VIV on the Humen Bridge resulted in a nine-day traffic interruption, leading to considerable economic losses and drawing widespread public attention to the issue of VIV (Ge et al., 2022; Zhao et al., 2022). Therefore, it is crucial to suppress VIV of existing bridges (in-service or under-construction).

The VIV of bridges is affected by many parameters, such as structural damping, aerodynamic shapes and wind turbulences. Among them, the structural damping is the most important parameter. For example, the bridges that experienced VIV including the Humen Bridge (Ge et al., 2022; Zhao et al., 2022), the Second Severn Bridge (MacDonald et al., 2002), the Yingwuzhou Yangtze River Bridge (Cao et al., 2023) and the Storebælt Suspension Bridge (Larsen et al., 2000) all exhibit relatively low damping. Additionally, a poor aerodynamic shape significantly contributes to VIV. For example, changes in aerodynamic shape caused by water-filled barriers and the maintenance tracks directly triggered the VIV of the Humen Bridge (Ge et al., 2022). Similarly, the VIV of the Yi Sun-sin Bridge is attributed to the improper design of the blocked guardrails (Hwang et al., 2020). Moreover, some perspectives suggest that wind turbulence can also influence the WIVs of bridges (Barni and

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Mannini, 2024; Li et al., 2024; Yu et al., 2022); however, the precise impact remains a topic of controversy (Hui et al., 2024; Xie et al., 2024; Zhu et al., 2024).

VIV mitigation measures of bridges can generally be categorized into three types: structural, mechanical and aerodynamic measures. For long-span bridges, mechanical and aerodynamic measures are more commonly implemented in practical engineering applications. Among mechanical measures, dampers are the most widely used (Elias and Matsagar, 2017). Examples include tuned mass dampers (TMD) installed on the Tokyo Bay Bridge and the Humen Bridge (Fujino and Yoshida, 2002; Ge et al., 2022; Zhao et al., 2022), as well as eddy current dampers on the Yingwuzhou Yangtze River Bridge (Cao et al., 2023; Hua et al., 2019). Aerodynamic measures are also extensively employed to mitigate VIV, largely due to their simplicity and reliability in installation. Common aerodynamic components include flow-suppressing plates, wind fairings, lower stability plates, horizontal wing plates and guide vanes, which have been installed on various bridges (Dong et al., 2021; Ge et al., 2022; Li et al., 2018, 2019; Liu et al., 2023a, 2023b; Yang et al., 2021; Larsen et al., 2000).

In addition to conventional two-dimensional aerodynamic measures, three-dimensional geometric modifications have also been validated effective in mitigating the VIV (Choi et al., 2008). The aerodynamic mechanism behind these measures lies in their ability to trigger three-dimensional wake instability of bluff bodies (Doddipatla et al., 2017; Williamson, 1996a, 1996b). This approach has been widely adopted in the engineering field, and Gao et al. (2021) provided a detailed introduction in their review. A notable case study is by El-Gammal et al. (2007), who successfully mitigate the VIV of a bridge girder by spanwise periodic perturbation method (SPPM). Additionally, measures such as partially blocking guardrails (Bai et al., 2020), installing wavy railings (Zhan et al., 2020), and incorporating small wind turbines (Zhang et al., 2022) can also

be regarded as three-dimensional geometric modifications, which can effectively suppress VIV. Vortex generators (VG) are also a type of three-dimensional geometric modification frequently employed on aircraft wings to delay boundary layer separation. They have applications in various fields, including aerospace, automotive, and railway industries. (Li et al., 2023; Lin, 2002; Yan et al., 2011). At low wind speeds, VGs can generate streamwise vortices, disrupt the existing large-scale vortical structures, and trigger three-dimensional wake instability. Due to its special aerodynamic characteristics, VGs are considered capable of mitigating VIV of bridges. Xin et al. (2018) validate the VIV mitigation effect of VGs on a bridge girder through wind tunnel tests, while Zhang et al. (2019) investigated their effectiveness and underlying mechanisms using numerical simulations. However, despite the wind tunnel tests and numerical simulations demonstrating the effectiveness of VGs, there is currently no evidence indicating their feasibility on real bridges. The mitigation effect of VGs on VIV needs to be tested through practical applications.

This paper presents a case study of VIV investigation and control of an in-service cable-stayed bridge with steel-concrete composite girder, which experienced unexpected WIV under a common wind speed (about 10 m/s) in 2023. On-site inspection, field measurement, wind tunnel tests and CFD simulation were conducted to investigate the cause of the unexpected WIV, and to find out a solution of it. An aerodynamic solution based on VGs was proposed and practiced, and the efficiency was verified based on long-term monitoring.

2. Unexpected wind-induced vibration of the bridge

2.1. Engineering background

The bridge depicted in Fig. 1 is a cable-stayed bridge in China with a main span of 460 m adopting Π -shaped steel-concrete composite girder,

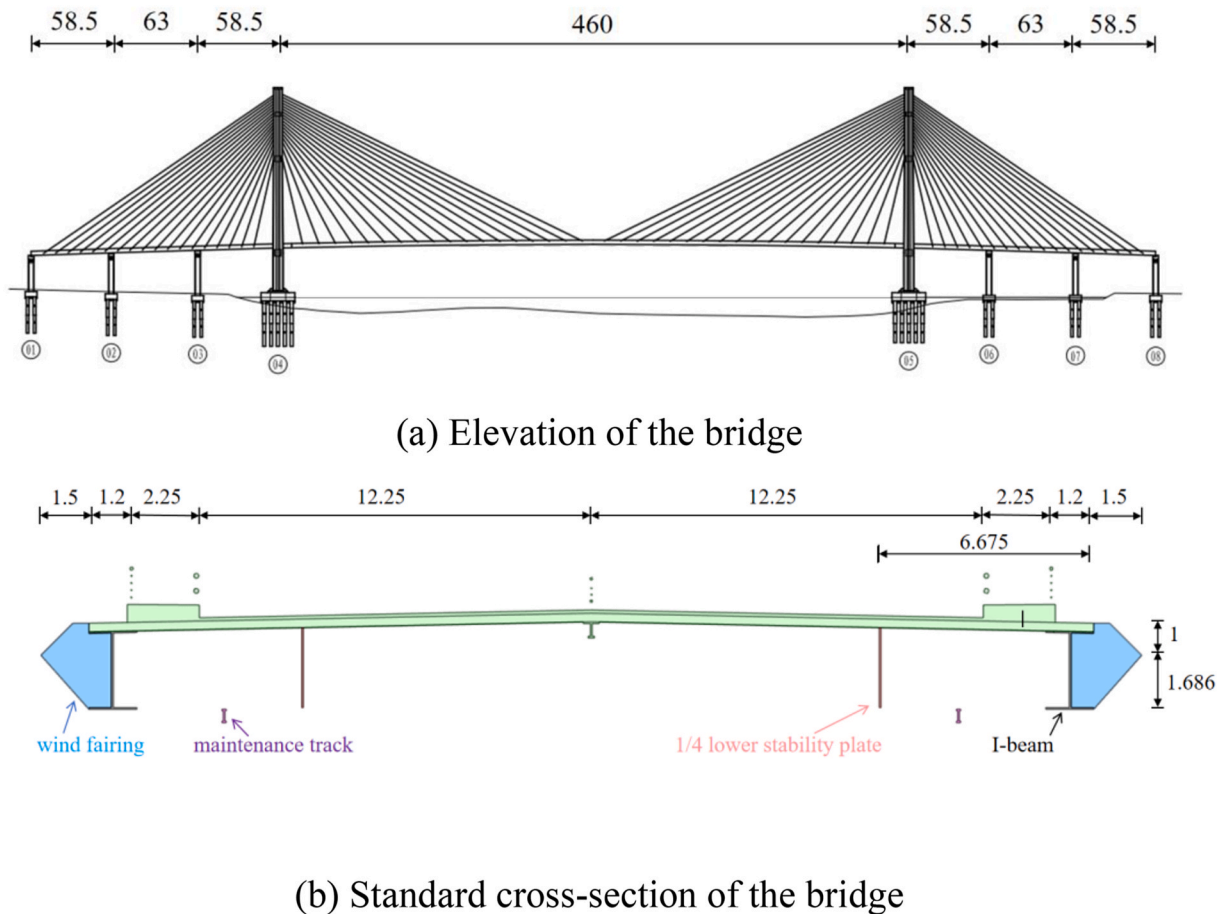


Fig. 1. Overall layout of the bridge in China (Unit: m).

which was completed in April 2014. The main girder of the bridge is 34.4 m wide (including the wind fairings) and 3.1 m high. Two 2.5m-high lower stability plates are installed at the bottom, and wind fairings are installed on both sides of the girder. Besides, the bridge adopts a semi-floating system with vertical supports, transverse wind-resistant supports and longitudinal dampers set in the tower.

2.2. Occurrence of the unexpected wind-induced vibration

On May 6, 2023, significant unexpected WIV of the bridge girder was observed and reported for the first time. Due to the sudden occurrence and the absence of a health monitoring system at that time, no vibration data was recorded. However, visual observations indicated that the main girder experienced vertical vibrations with an amplitude of approximately 10 cm, which was significantly larger than a common amplitude under normal weather conditions (see Fig. 2). In the following months, similar unexpected WIVs occurred multiple times under the same wind speed. These incidents disturbed traffic and caused discomfort of pedestrians and divers on the bridge deck. Therefore, there is an urgent need to address this problem.

There were no reported instances of unexpected large-amplitude WIVs in the bridge's service history (without a health monitoring system) prior to the event observed on May 6, 2023. However, it is possible that such unexpected WIVs had occurred earlier but went undetected or unnoticed due to the absence of a health monitoring system. Hence, to investigate the unexpected WIVs, a health monitoring system was installed on the bridge in July, which continuously collected the vibration response and wind speed data. Preliminary analysis of the measured data from July raised concerns that the unexpected WIV observed before was VIV. In August, the health monitoring system was upgraded to include intelligent early warning capabilities, resulting in the collection of more valuable data, which is summarized in Table 1 for reference. The data indicated that as unexpected WIVs occurred, the wind speed was around 10 m/s, and the peak acceleration of the girder at the midspan reached about 50 cm/s² (see Fig. 3). It is worth noting that the vibration frequency varies slightly with wind speed within the lock-in range.

The acceleration time history data collected by the health monitoring system were subjected to Fourier transform, and the resulting vibration frequency closely matches the frequency of the first-order symmetrical vertical bending mode. This confirms that the first-order symmetric vertical bending mode is the primary mode excited by the wind.

Table 1

Some unexpected WIV responses recorded in the summer of 2023.

Date	Peak acceleration at midspan (cm/s ²)	Vibration Frequency (Hz)	Mean Wind speed (m/s)
2023.08.07	28.4	0.33–0.34	9.5–11.0
2023.08.08	27.6		
2023.08.09	19.1		
2023.08.10	30.4		
2023.08.11	26.6		
2023.08.13	50.0		
2023.08.14	32.5		
2023.08.15	24.2		
2023.08.26	22.4		
2023.08.27	32.9		
2023.09.12	39.8		

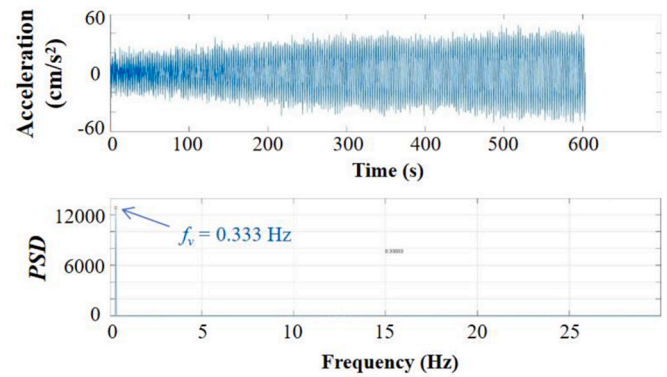


Fig. 3. Unexpected WIV responses of the bridge (Mean Wind speed $U = 9.98$ m/s).

2.3. Preliminarily supposed causes of the unexpected wind-induced vibration

After the unexpected WIV being observed for the first time, China Merchants Chongqing Communications Technology Research & Design Institute Co., LTD, Hunan University and Southwest Jiaotong University was immediately involved in the investigation of it. According to the initial investigation, there were two preliminarily supposed causes of the unexpected WIVs: the change of central guardrail and the low structural damping. Subsequently, we confirmed that the change of central guardrail was not the cause of the unexpected large-amplitude



Fig. 2. Unexpected WIV of the bridge.

WIVs, but the low structural damping (significantly lower than the assumed values of the recognized codes) was a key trigger factor for the unexpected WIVs. This will be analyzed in detail below.

2.3.1. Change of the central guardrail

Any change of the aerodynamic shape of the bridge may cause large-amplitude VIV, such as the blocked guardrails of the Yi Sun-sin Bridge (Hwang et al., 2020) and the maintenance track and the water-filled barriers of the Humen Bridge (Ge et al., 2022). As a critical auxiliary structure of the bridge, guardrails can alter the aerodynamic performance of the main girder. Studies by Nagao et al. (1997) have shown that the size and position of the horizontal bars on guardrails can significantly affect vibration amplitude. Matsumoto et al. (1993) demonstrated that the amplitude of VIV varies with the porosity of the guardrails. Moreover, changing the overall configuration of guardrails can sometimes mitigate VIVs. For instance, Xin et al. (2021) and Zhan et al. (2020) successfully mitigated VIV of the bridge deck by using innovative designs like inclined and wavy railings, respectively. And Bai et al. (2020) found that partially blocked guardrails were highly effective in mitigating VIVs.

Interestingly, the bridge had just undergone a change of central guardrail in March (about two months prior to the first occurrence of unexpected WIV). The before-and-after configurations of the central guardrails are shown in Fig. 4. As can be seen, the aerodynamic shape changed significantly, which likely altered the aerodynamic characteristics of the main girder and may have exacerbated the unexpected WIVs.

However, it is worth noting that the test results discussed below indicate that changing the central guardrail has little effect on the VIV performance of the girder, thereby ruling out this factor as a preliminarily supposed cause.

2.3.2. Low structural damping

The bridge had been operational for over nine years prior to the first report of unexpected WIV. The dynamic characteristics of the bridge were estimated based on finite element analysis of full-bridge during the design phase before construction. And the structural damping of the bridge was determined through field measurement using acceleration sensors under ambient excitations. It is worth to note that during the field measurements, the bridge was still in service, and acceleration sensors were only installed on one side of the bridge deck. On the other

hand, unexpected WIV was only observed on the first symmetric vertical mode of the bridge girder. Consequently, measurements of the damping for the torsional mode were not conducted. The data collected by the acceleration sensors were analyzed through the Stochastic Subspace Identification (SSI) method and the Analytical Mode Decomposition (AMD) method respectively (Magalhaes et al., 2007; Viviescas Jaimes et al., 2018). All results are presented in Table 2.

As shown in Table 2, the vibration frequency obtained from field measurements exhibit discrepancies of less than 15% compared to those derived from finite element analysis, indicating a relatively high level of accuracy. The damping ratios identified through both the SSI method and the AMD method are all at low levels, differing significantly from the assumed values of the recognized design codes (for bridges with steel-concrete composite girder, the recommended structural damping in Europe code is 0.0064, while in China code it is 0.01) (European Commission, 2005; The People’s Ministry of Transport of China, 2019). The measured damping was adopted in the wind tunnel tests in Section 3.2.3.

Table 2
Structural dynamic characteristics of the bridge.

Mode	Vibration frequency (Hz)		Damping (%)	
	Finite element analysis	Field measurement	SSI-COV method	AMD-RDT method
First-order symmetrical vertical	0.317	0.339	0.20	0.25
First-order anti-symmetrical vertical	0.406	0.434	0.43	0.32
Second-order symmetrical vertical	0.567	0.611	0.35	0.41
Second-order anti-symmetrical vertical	0.673	0.748	0.37	0.41
First-order symmetrical torsional	0.604	0.614	–	–
First-order symmetrical torsional	0.730	0.812	–	–



(a) The central guardrail in current situation (Central Guardrail 1, 2023.03 - now)



(b) The central guardrail in the past (Central Guardrail 2, 2014.04 - 2023.03)



Fig. 4. Change of the central guardrail.

Structural damping is a crucial parameter that influences the VIV performance of bridges. However, the structural damping of long-span bridges with steel-concrete composite girder exhibits considerable variability and is challenging to predict accurately during the design phase. For example, the vertical damping ratio of Second Severn Bridge is as low as 0.29%, and the Yingwuzhou Yangtze River Bridge is 0.25% (Cao et al., 2023; MacDonald et al., 2002). Also, as shown in Table 2, the present results of damping based on field measurements on the bridge was as low as 0.2%–0.4%. All above results differed significantly from the assumed values of the recognized design code. It is speculated that the low structural damping might be an important reason for the VIV of the bridge (this will be analyzed in detail in Section 3.2.3).

3. Experimental study on VIV performance

3.1. Experimental setup

The universal wind tunnel tests using an elastically-mounted sectional model were employed to examine the VIV performance of

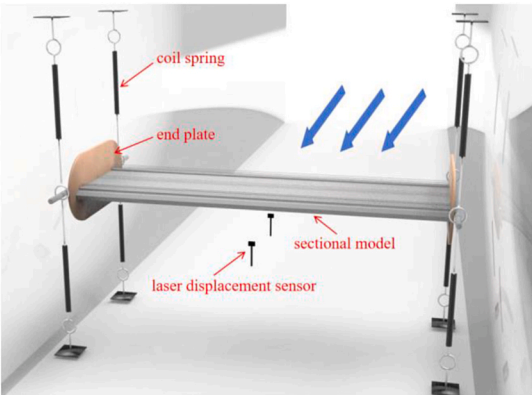
the main girder (Diana et al., 2013; Meng et al., 2023, 2024; Wang et al., 2021, 2022b). To fully account for the Reynolds number effect, two sectional models with scale ratios of 1:50 and 1:25 were used (Li et al., 2014a,b; Raghavan and Bernitsas, 2011; Ran et al., 2024; Schewe and Larsen, 1998). The tests were carried out in the high- and low-speed sections of the HD-2 wind tunnel at Hunan University. The high-speed section measures 17 m in length, 3 m in width, and 2.5 m in height, with adjustable wind speeds ranging from 0 m/s to 58 m/s. The low-speed section, an open section, measures 15 m in length, 8.5 m in width, and 2.0 m in height, with wind speeds adjustable from 0 m/s to 18 m/s. In both sections, the incoming flows were stable and uniform throughout the study.

The sectional models were made of plastic plates, wood, and two steel core girders. To minimize end effects, the ends of the sectional models were capped with wooden plates (Wang et al., 2022a). Table 3 provides the geometrical parameters of the two sectional models.

Additionally, a suspension system was utilized to test the free vibration characteristics of the sectional models. As depicted in Fig. 5, the models were suspended within the wind tunnel using coil springs

Table 3
Parameters of spring-mounted suspension system for two girders.

Parameters	Full-scale bridge	Sectional model	
Geometric scale ratio λ_g	—	1:50	1:25
Height H (m)	3.0	0.06	0.12
Width B (m)	34.4	0.688	1.376
Length L (m)	—	2	4
Aspect ratio L/B	—	2.91:1	2.91:1
Wind speed ratio	—	1:3.85	1:3.65
Mass m (Kg/m)	50179.2	20.06	80.29
Mass moment of inertia J (Kg•m ² /m)	4886450	0.782	12.509
Vertical frequency f_v (Hz)	0.339	4.12	2.18
Torsional frequency f_α (Hz)	0.614	7.86	4.14
Structural damping ratio ξ_v	—	0.35%–0.80%	0.33%
Structural damping ratio ξ_α	—	0.44%–0.94%	0.32%



(a) Diagram of spring-suspended system



(b) 1:50 sectional model



(c) 1:25 sectional model

Fig. 5. The spring-suspended systems of sectional models.

attached to the walls. The wind-induced vibrations were recorded by two laser displacement meters positioned beneath the model, operating at a sampling frequency of 1000 Hz. Wind speed was measured by an ultrasonic anemometer placed at a suitable distance from the sectional model on the windward side. During each test, careful measurements were taken at every test point, and for each wind speed, 30 s of data were recorded after reaching a steady state.

3.2. VIV performance of the bridge girder

3.2.1. VIV response of the current girder

In consideration of the potential Reynolds number effect, two sectional models with different scale ratios were employed to investigate the VIV performance of the current girder (the girder installing Central Guardrail 1). In the wind tunnel tests, the VIV responses of the girder were recorded across various wind angles of attack (AoA) and wind speeds. The results for both vertical and torsional VIV responses are presented in in Figs. 6–7, where A_v/H denotes the dimensionless vertical vibration amplitude, A_α denotes the torsional vibration amplitude, and $U/f_v H$ denotes the reduced wind speed.

VIV responses depicted in Fig. 6 shows the test results of the 1:50 sectional model. Despite the inclusion of wind fairings and 1/4 lower stability plates as aerodynamic measures, the bridge still exhibited

large-amplitude VIVs at AoAs of 0° , -3° , and -5° under a low damping condition ($\xi_v = 0.35\%$, $\xi_\alpha = 0.44\%$). At the AoA of -5° , there were three distinct lock-in ranges: two for vertical VIV ($8.73 < U/f_v H < 10.59$ and $14.92 < U/f_v H < 20.82$) and one for torsional VIV ($18.49 < U/f_v H < 25.40$). The second vertical lock-in range was absent at -3° AoA, while only the first vertical lock-in range persisted at 0° AoA. The most terrible AoA was -5° , where the maximum vertical and torsional VIV amplitudes reached $A_v/H = 0.0629$ and $A_\alpha = 0.55^\circ$, respectively.

VIV responses depicted in Fig. 7 show the test results of the 1:25 sectional model. As you can see, the VIV response pattern of the 1:25 sectional model exhibits similar characteristics, with a comparable number of VIV lock-in ranges and wind speeds at wind AoAs of -5° , -3° , and 0° , compared to the 1:50 sectional model. The maximum vertical and torsional VIV amplitudes were recorded at wind AoAs of -5° and -3° , with values of $A_v/H = 0.1155$ and $A_\alpha = 0.51^\circ$, respectively. The difference is that, at the -3° wind AoA, the VIV lock-in range increased from one to two distinct ranges. The differences observed in the test results between the two sectional models may be attributed to the following factors: (1) Reynolds number effects; (2) Turbulence intensity of wind tunnels (0.5% for the high-speed section and 1% for the low-speed section).

In addition, the test results provide evidence that the unexpected VIV of the actual bridge were successfully replicated in the wind tunnel

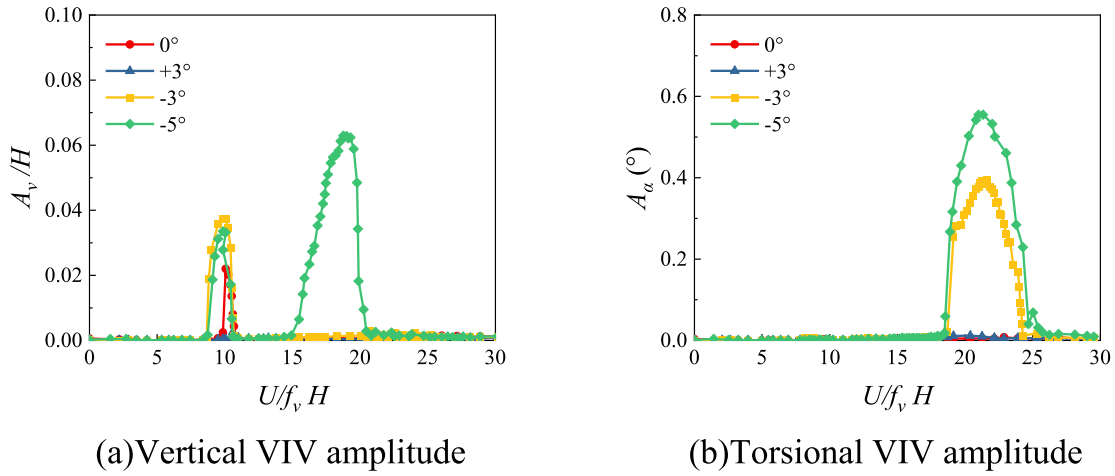


Fig. 6. The wind tunnel test result of the current girder (1:50 sectional model).

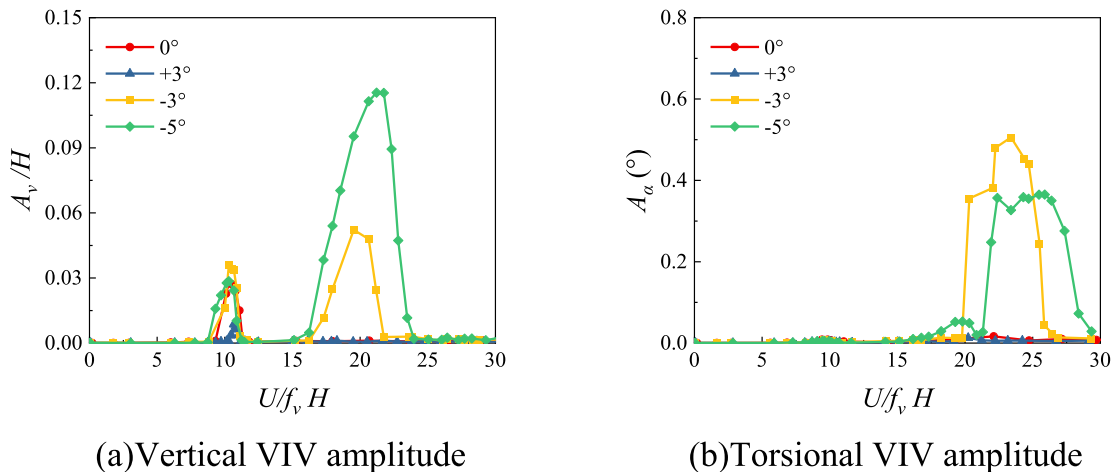


Fig. 7. The wind tunnel test result of the current girder (1:25 sectional model).

tests. For the 1:50 sectional model, the maximum VIV amplitude in the first vertical lock-in range at -3° AoA ($U/f_v H = 9.87$ and $A_v/H = 0.0335$) and -5° AoA ($U/f_v H = 9.87$ and $A_v/H = 0.0376$) closely align with the observed vibration amplitude ($U \approx 10$ m/s and $A_v \approx 0.1$ m). For the 1:25 sectional model, the maximum VIV amplitude in the first vertical lock-in range at -3° AoA ($U/f_v H = 10.30$ and $A_v/H = 0.0360$) and -5° AoA ($U/f_v H = 10.30$ and $A_v/H = 0.0285$) also shows strong consistency with the observed vibration amplitude.

3.2.2. Effect of changing the central guardrail

Nearly before the first report of the unexpected WIV, the central guardrail had already been changed (see Section 2.3.1 and Fig. 4). The wind tunnel tests were conducted using 1:50 sectional model to assess the impact of changing the central guardrail on the VIV performance of the girder. Fig. 4 illustrates the VIV responses of the girder with Central Guardrail 1, while Fig. 6 shows the VIV responses of the girder with Central Guardrail 2.

A comparison of the test results in Figs. 6 and 8 shows that there is almost no change in the VIV responses of the girder before and after changing the central guardrails. In fact, the VIV responses were slightly mitigated following the guardrail changing.

Overall, this suggests that changes to the central guardrail are unlikely to be the cause of the unexpected WIVs. This may be related to the high porosity of the central guardrail.

3.2.3. Effect of damping

Low structural damping has also been suspected as a contributing factor to unexpected WIVs. Therefore, detailed analyses of its impact were conducted using wind tunnel tests with the 1:50 sectional model. Without loss of generality, all tests utilized the sectional model equipped with Central Guardrail 1 and were conducted at the most terrible wind AoA of -5° .

Structural damping was identified using the Logarithmic Least Squares Fit (LLSF) method (Hua et al., 2020; Tang et al., 2019). This approach divides the displacement signals into multiple time windows, allowing for the identification of structural damping variations with amplitude through the movement of the time window. This method effectively captures the nonlinear characteristics of structural damping as it changes with amplitude. Furthermore, changes in structural damping in the tests were achieved by adjusting the length of electrical tape wound around the coil springs.

Fig. 9 illustrates the variations in vertical and torsional amplitudes as structural damping changes. As can be seen, the bridge will experience large-amplitude VIV under low damping ($\xi_v = 0.35\%$, $\xi_a = 0.44\%$). Both

vertical and torsional VIV amplitudes decrease significantly with higher damping ratios. When the vertical and torsional damping increase from 0.35% to 0.55% and from 0.44% to 0.65%, the vertical and torsional amplitudes decrease by 39% and 69%, respectively. With a further increase to 0.80% in the vertical damping and 0.94% in the torsional damping, the vertical amplitudes decrease by 65%, and the torsional VIV is completely eliminated.

Based on the test results, it is reasonable to conclude that low structural damping is an important factor contributing to the unexpected large-amplitude WIVs of the bridge. However, it is worth noting that low damping is an important but possible not the sole factor. There is no evidence to suggest whether unexpected WIVs had occurred previously, nor is there any evidence that the damping significantly decreased prior to the first notice VIV on May 6, 2023. In another similar case of the Yingwuzhou Yangtze River Bridge with a steel-concrete girder, on which VIV has occurred suddenly in 2020 after 6 years of opening of the bridge, the structural damping of the bridge girder was also found to be very low, which is quite similar to the situation in our study (Cao et al., 2023). However, due to the lack of long-term monitoring data, it is difficult to determine whether the damping decreased over time or if it was always low from the opening of the bridge.

Besides, changes of the wind environment may also be a potential cause of the occurrence of large-amplitude VIV (Han et al., 2022; Li et al., 2014a,b; Ma et al., 2020; Zhao et al., 2022). For instance, the Yingwuzhou Yangtze River Bridge (with a steel-concrete girder), situated in the monsoon region of the middle Yangtze River, had experienced significant variations in wind environment with different seasons and years, which may have been a potential cause to the large-amplitude VIV. Similarly, the bridge in our study is located in a coastal region, where the wind environment is even more complex and variable. Thus, it is likely that changes in the wind environment played a role in the onset of large-amplitude VIV.

In conclusion, we believe that two potential causes may have contributed to the sudden onset of large-amplitude VIV: (1) reductions in structural damping during the bridge's service period (2) changes in the local wind environment. However, due to the lack of long-term monitoring data on both the damping and wind environment, it is difficult to draw definitive conclusions now. In the future, it is worth to conduct long-term monitoring and experimental research throughout the entire operational life of the bridge.

Besides, due to the II-shaped girder and I-shaped edge girders, there is limited space for installing additional tuned mass dampers (TMD). Therefore, aerodynamic solutions have become the preferred choice for mitigating the unexpected WIVs.

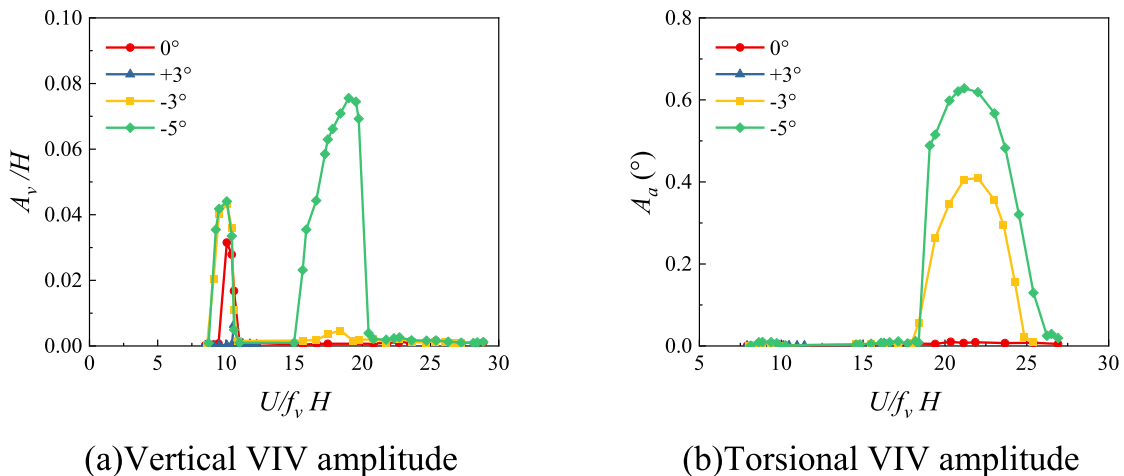


Fig. 8. The wind tunnel test result of the 1:50 section model installing guardrail 2.

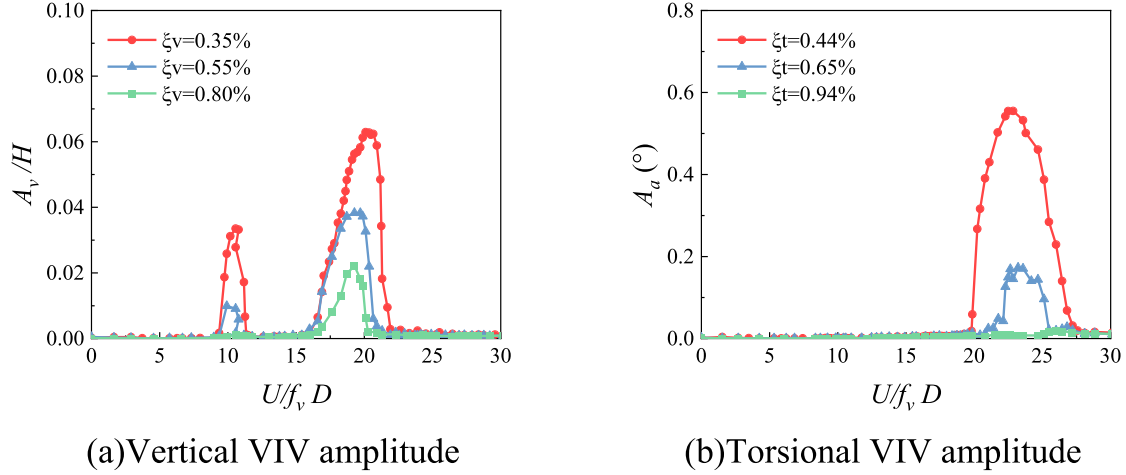


Fig. 9. The wind tunnel test results of the VIV effect of the girder under different damping ratios at the -5° AoA.

3.3. Aerodynamic solutions

Aerodynamic measures are a commonly used method for VIV mitigation (Fujino and Siringoringo, 2013). In this study, the VIV mitigation effects of dozens of conventional aerodynamic measures and a new aerodynamic measure are tested in depth.

3.3.1. Conventional aerodynamic measures

Dozens of conventional aerodynamic measures such as wind fairings, flow-suppressing plates, horizontal wing plates and guide vanes are investigated on the 1:50 sectional model (Dong et al., 2021; Ge et al., 2022; Larsen et al., 2000; Li et al., 2019). But due to the heavy workload, details of the VIV responses of many cases are not recorded. Some of the important aerodynamic measures are shown in Fig. 9 and Table 4. The vertical and torsional damping were set at 0.35% and 0.44%, respectively, with a test AoA of -3° . Additionally, all parameters of the sectional model and test settings are identical to those described in Section 3.1.

Figs. 11–14 provide a detailed presentation of the wind tunnel test results for the conventional aerodynamic measures described in Fig. 10 and Table 4. It is evident that, under low structural damping, most conventional aerodynamic measures exhibit weak mitigation effect of VIVs and may even exacerbate VIV in some cases. Hence, these measures were quickly ruled out. (see Fig. 12, Fig. 13)

However, it is worth mentioning that the horizontal wing plate demonstrated better VIV mitigation effects compared to most other measures. As shown in Fig. 14, with the 0.9m horizontal wing plate,

VIVs were observed only in the first vertical lock-in range. When the length of the horizontal wing plate was increased to 1.5m, the VIV in the first lock-in range were reduced to half of those recorded with the current girder. Despite its favorable mitigation performance, this aerodynamic measure was ultimately rejected due to high cost and impact on the overall aesthetics of the bridge.

Above all, the bridge's VIVs is challenging to fully be mitigated using conventional aerodynamic measures, indicating the need for new approaches. Although we did not identify a suitable aerodynamic solution, we found that aerodynamic measures installed on the upper side of the girder had virtually no effect on VIV suppression, whereas those installed on the underside showed some mitigating effects. This suggests that the aerodynamic cause of the bridge's VIVs may originate from the lower edge of the bridge.

3.3.2. Vortex generators

Under low damping, it proved challenging to identify a suitable conventional aerodynamic measure for VIV mitigation at -3° AoA. Consequently, we opted to test the VIV mitigation effect of a novel aerodynamic measure: vortex generators (VG). VGs were initially developed in the aerospace industry to delay boundary layer separation (Lin, 2002). In recent years, some researchers have introduced VGs into the field of bridge wind engineering, investigating their effectiveness in suppressing VIVs of bridges through wind tunnel tests and computational fluid dynamics (CFD), yielding good results (Tao et al., 2016; Xin et al., 2018; Zhang et al., 2019).

In this test, the vortex generators (VGs) were mounted on the lower side of the wind fairings, as shown in Fig. 15. Each VG consists of two plates, inclined in opposite directions and forming a 45° angle with the bottom edge. Two key parameters influence the effectiveness of VGs in suppressing VIVs: height and spanwise distance. Therefore, six different heights ($h = 0.13H, 0.15H, 0.17H, 0.2H, 0.27H$ and $0.3H$) and two different spanwise distances ($s = 2.67H$ and $2H$) are considered in the tests.

Initial wind tunnel tests are conducted on the 1:50 sectional model, with all test settings consistent with those in Section 3.1. The vertical and torsional damping are selected 0.35% and 0.44%, respectively, and the AoA was maintained at -3° .

Fig. 16 presents a preliminary investigation into the VIV mitigation effects of VGs. As shown, increasing the height of the VGs (h) from 0.17H to 0.3H, or decreasing the spanwise distance (s) of the VGs from 2.67H to 2H, results in a reduction of the VIV amplitude. The most effective VIV mitigation occurs when the VG height is 0.17H and the spanwise distance is 2H.

Table 4
Conventional aerodynamic measures for VIV.

Measure	Notes of measure
I	No control
II	Blocked the central guardrails
III	Remove the maintenance tracks
IV	Lower central stability plate
V	Partially-blocked handrails
VI	Remove handrails
VII	Remove 1/4 lower stability plates
VIII	Heighten the 1/4 lower stability plates by 32 cm
IX	Flow-suppressing plates
X	1.2m inclined plate on the outer side of the sidewalk
XI	0.9m inclined guide vane at the bottom of the beam
XII	1.2m + 1.2m zigzag guide vane at the bottom of the beam
XIII	0.9m horizontal wing plate at the outer edge of the wind fairing
XIV	1.5m horizontal wing plate at the outer edge of the wind fairing

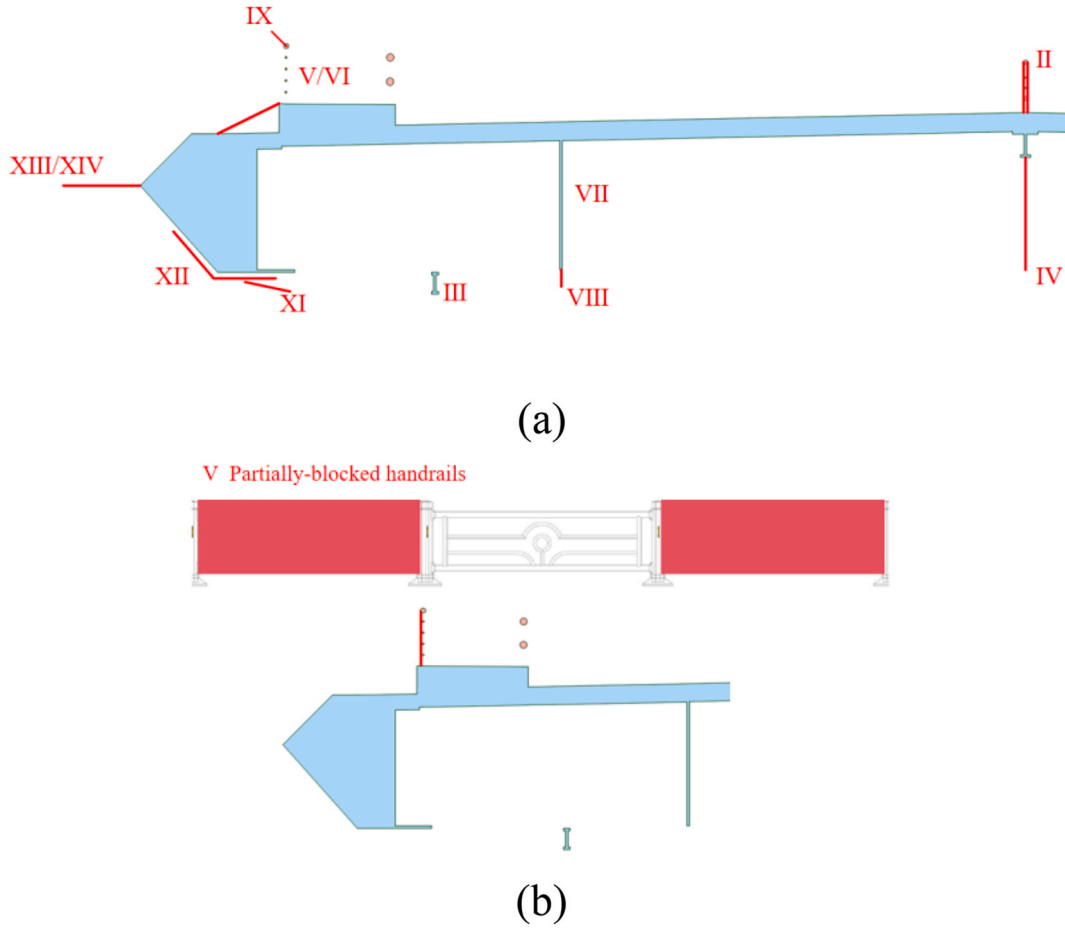
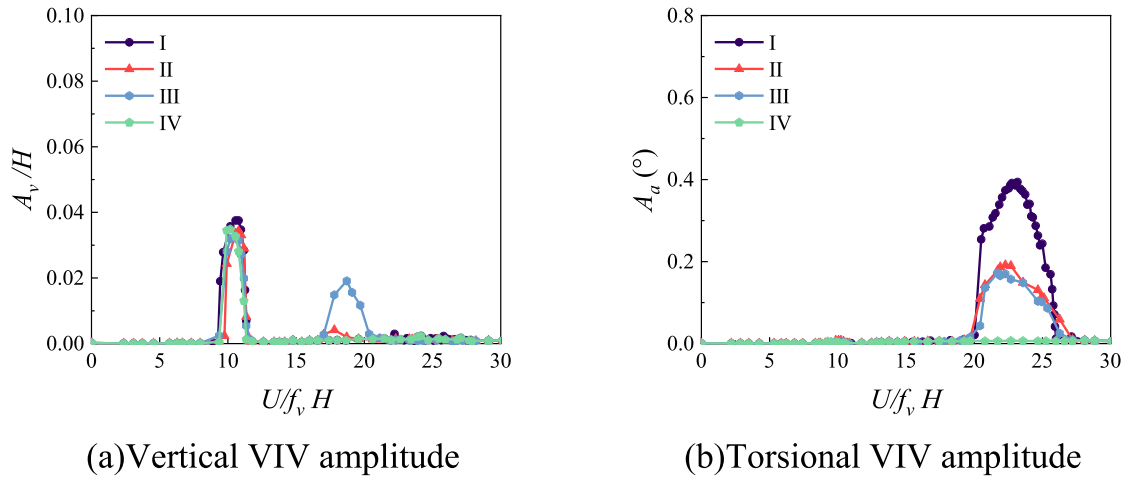


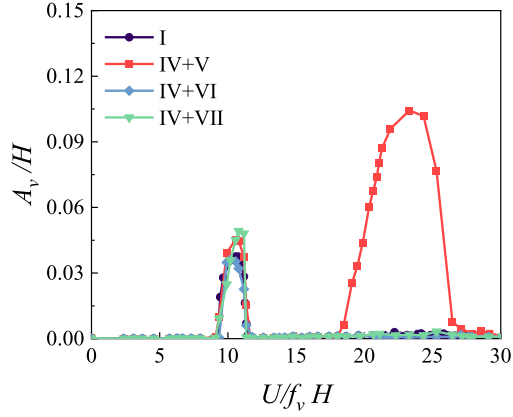
Fig. 10. Conventional aerodynamic measures.

Fig. 11. The wind tunnel test results of the VIV mitigation effect of Measure II,III and IV at the -3° AoA (1:50 sectional model).

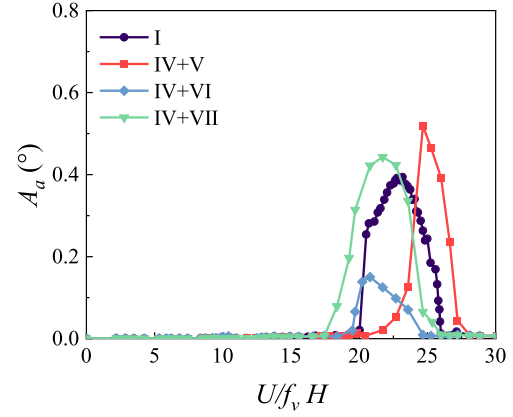
In the tests, we found that the VIV mitigation effect is highly sensitive to changes in the height of the VGs. Therefore, we maintained the spanwise distance of the VGs at $2H$ and conducted a new test to further explore the impact of variations in VG height. As shown in Fig. 17, when the height was $0.13H$, the vertical amplitude of VIV decreased from $0.0376H$ to $0.0290H$, representing a reduction of 23% compared to the current girder. As the VG height was further increased to $0.2H$ and

$0.27H$, the VIV was completely suppressed. Subsequently, the VIV mitigation effect of the $0.2H$ -high VG was examined at three additional AoAs (-3° , 0° , and $+3^\circ$). The results shown in Fig. 18 indicate that the VIV was also eliminated at these angles.

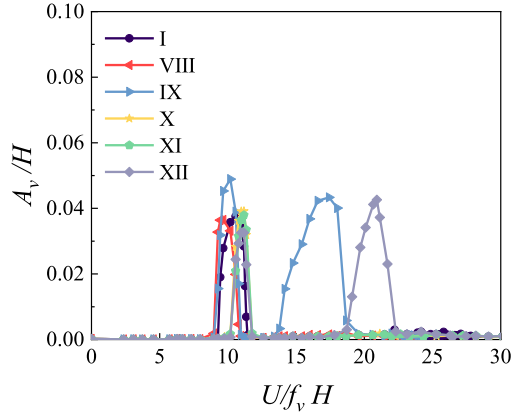
In consideration of the potential Reynolds number effect, the 1:25 sectional model was used to further verify the test results from the 1:50 sectional model, with results shown in Fig. 19. As illustrated, at the AoA



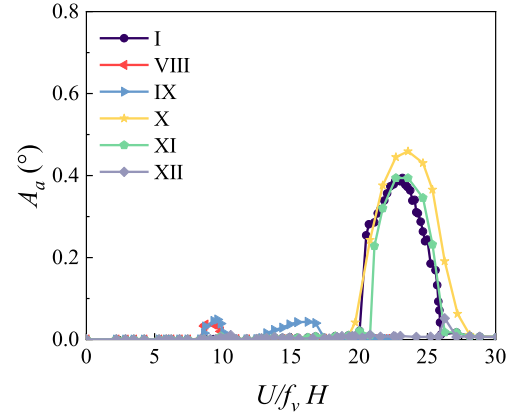
(a) Vertical VIV amplitude



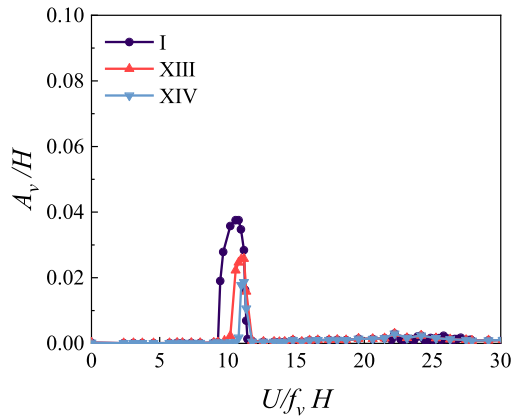
(b) Torsional VIV amplitude

Fig. 12. The wind tunnel test results of the VIV mitigation effect of the combination measures at the -3° AoA (1:50 sectional model).

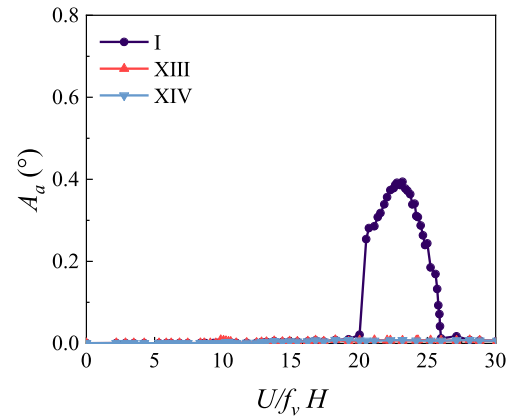
(a) Vertical VIV amplitude



(b) Torsional VIV amplitude

Fig. 13. The wind tunnel test results of the VIV mitigation effect of Measure VIII to XII at the -3° AoA (1:50 sectional model).

(a) Vertical VIV amplitude



(b) Torsional VIV amplitude

Fig. 14. The wind tunnel test results of the VIV mitigation effect of Measure XIII and XIV at the -3° AoA (1:50 sectional model).

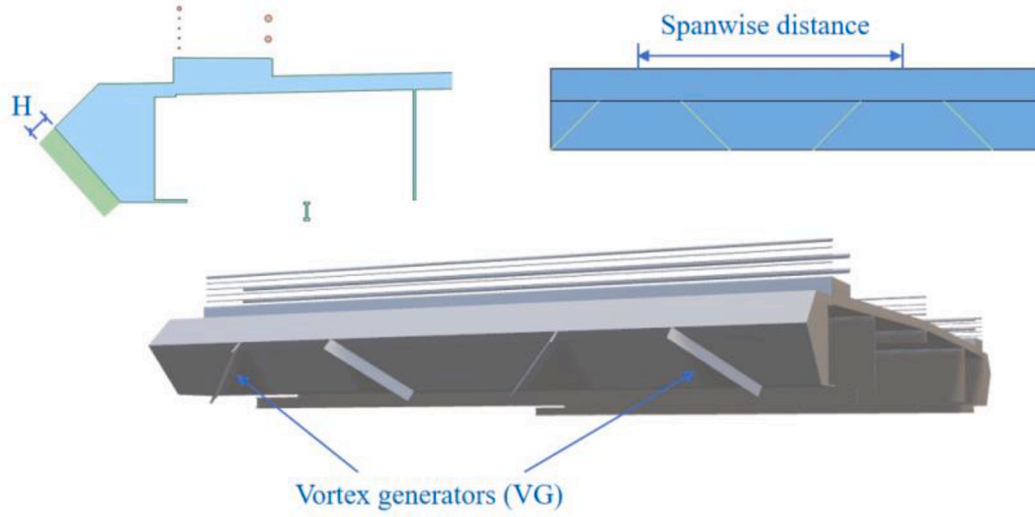
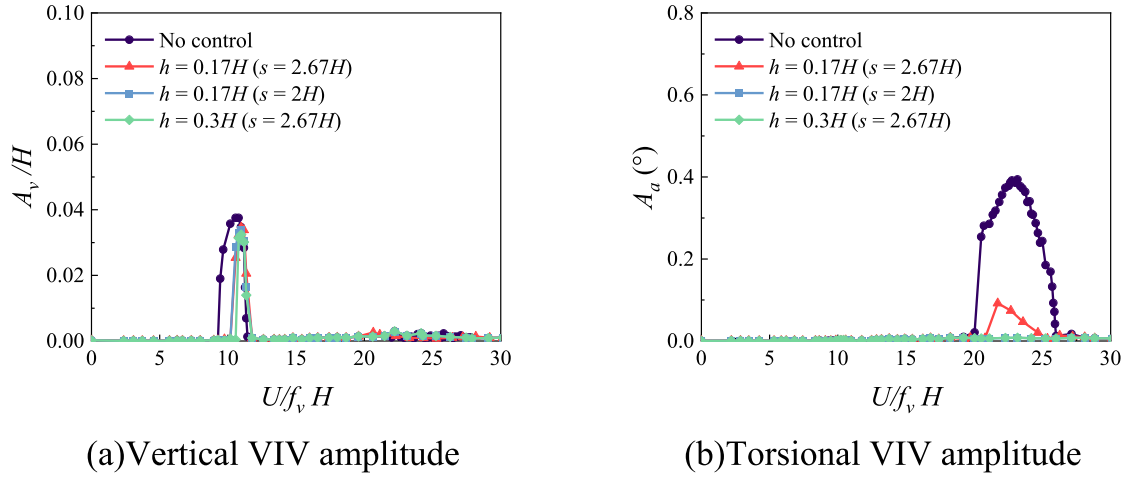
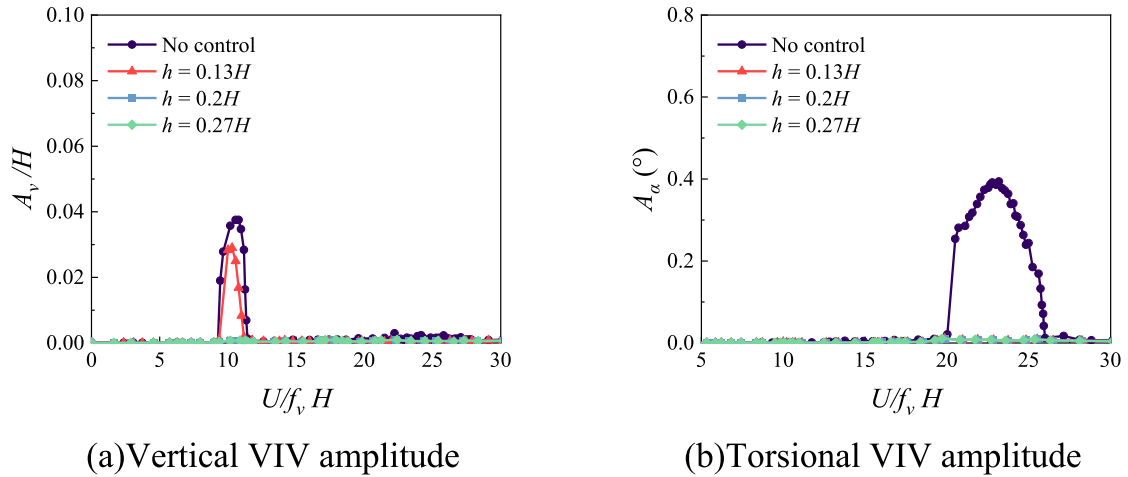
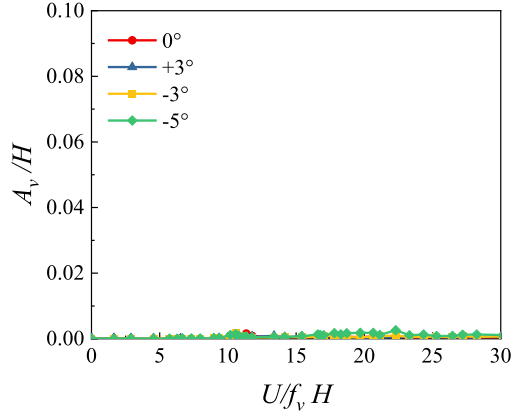
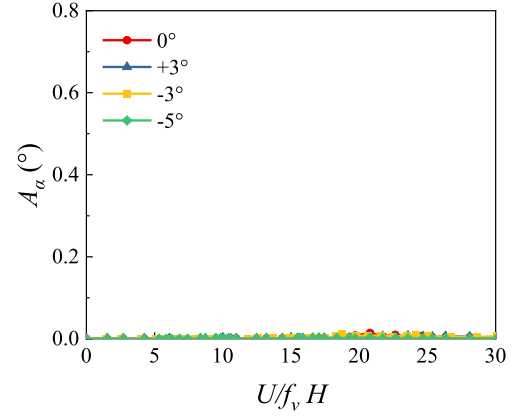


Fig. 15. Arrangement of VG

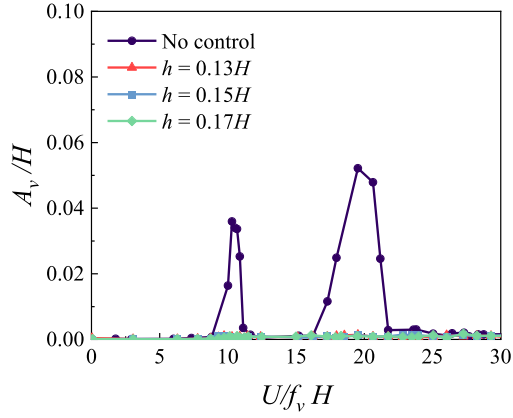
Fig. 16. The wind tunnel test results of the VIV mitigation effect of VG with different spanwise distances and heights at the -3° AoA (1:50 sectional model).Fig. 17. The wind tunnel test results of the VIV mitigation effect of VG with different heights at $s = 2H$ at the -3° AoA (1:50 sectional model).



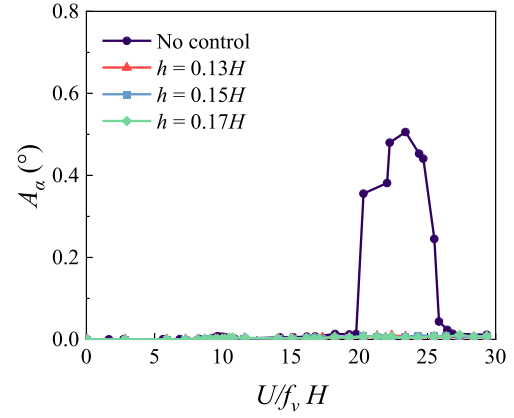
(a) Vertical VIV amplitude



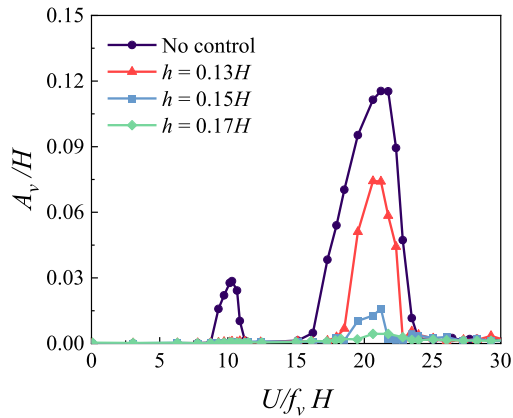
(b) Torsional VIV amplitude

Fig. 18. The wind tunnel test results of the VIV mitigation effect of VG with the height of $0.2H$ (1:50 sectional model).

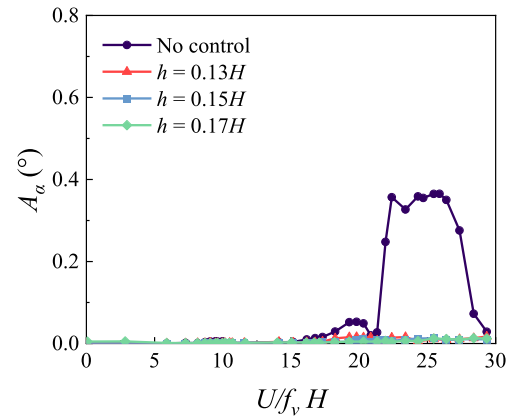
(a) Vertical VIV amplitude



(b) Torsional VIV amplitude

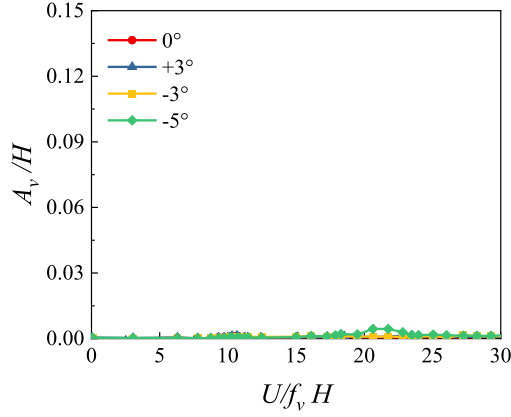
Fig. 19. The wind tunnel test results of the VIV mitigation effect of the VG with different heights at the -3° AoA (1:25 sectional model).

(a) Vertical VIV amplitude

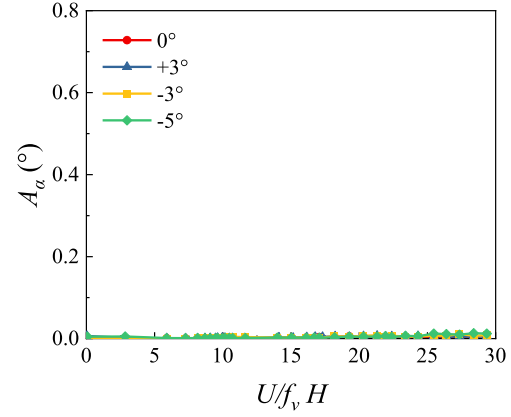


(b) Torsional VIV amplitude

Fig. 20. The wind tunnel test results of the VIV mitigation effect of the VG with different heights at the -5° AoA (1:25 sectional model).



(a) Vertical VIV amplitude



(b) Torsional VIV amplitude

Fig. 21. The wind tunnel test results of the VIV mitigation effect of the VG with the $0.17H$ height at four different AoA (1:25 sectional model).

of -3° , the VIV responses were entirely eliminated when VG heights were set to $0.13H$, $0.15H$, and $0.17H$. Fig. 20 presents the test results at -5° AoA, indicating that all three VG heights effectively mitigated torsional VIV, but only the $0.17H$ -high VG fully suppressed vertical VIV. Afterwards, the VIV mitigation effect of the $0.17H$ -high VG was tested at three additional AoAs, and the results indicate effective performance (see Fig. 21).

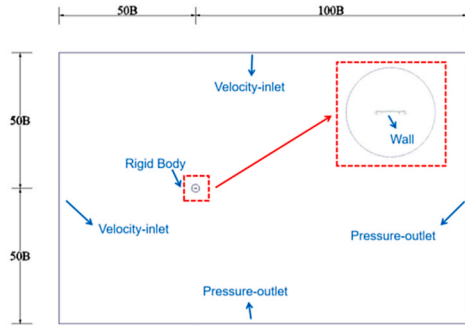
Above all, under low structural damping of about 0.35%, the VG with a height of $0.17H$ and a spanwise distance of $2H$ was able to completely mitigate the VIV. Finally, this highly effective aerodynamic measure was selected as the aerodynamic solution and was practiced on the actual bridge (see in Section 5).

4. CFD simulation

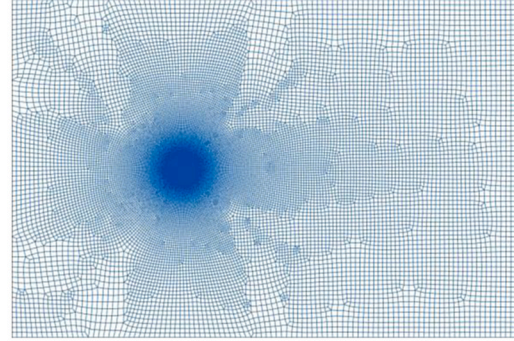
The computational fluid dynamic (CFD) method is widely used to simulate the real flow field for the most structures (Van Hooff et al., 2017) due to its low computational cost and fast processing speed. In this study, CFD simulations were employed to reveal the aerodynamic mechanism, complementing the wind tunnel tests.

4.1. Cases and models

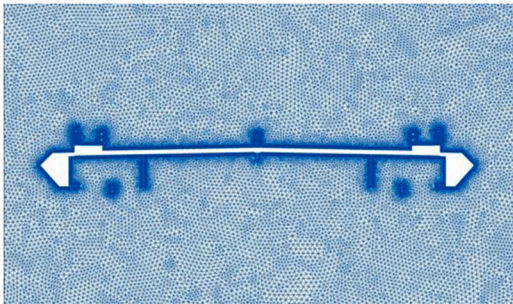
Two cases are selected for numerical simulation: (1) the current girder without VG. (2) the current girder with VG ($h = 0.17H$; $s = 0.2H$). As shown in Fig. 20, the fluid domain is a cuboid, where the height is $100B$, the width is $150B$, and the spanwise length is $8D$ (D and B is the



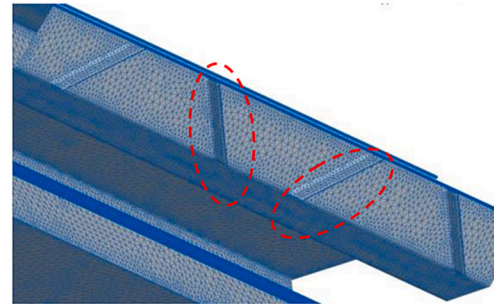
(a) Boundary conditions



(b) Global mesh

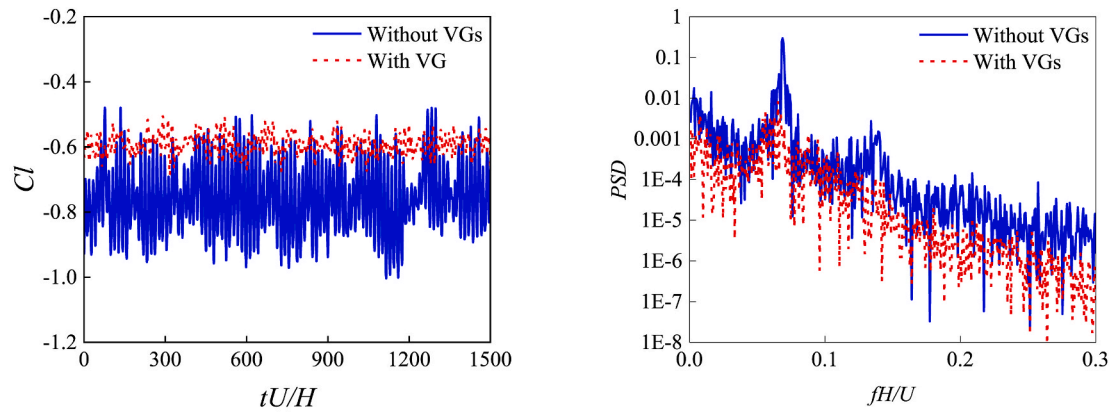
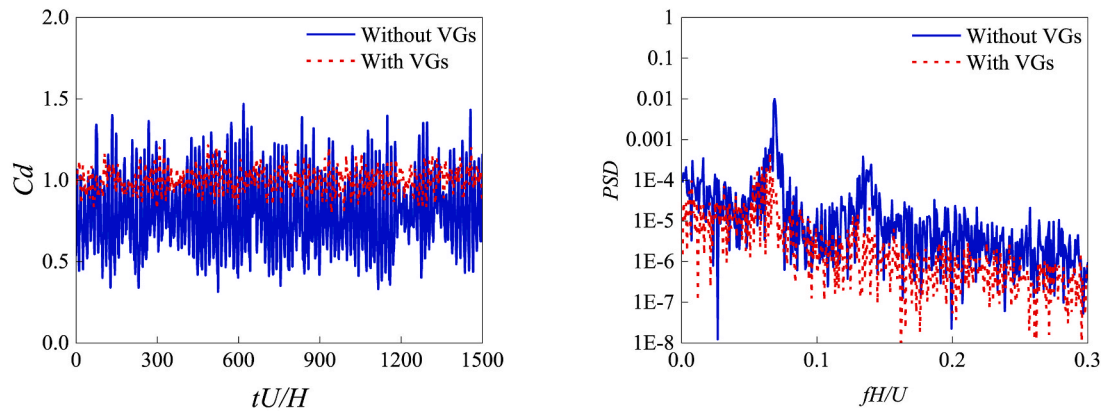
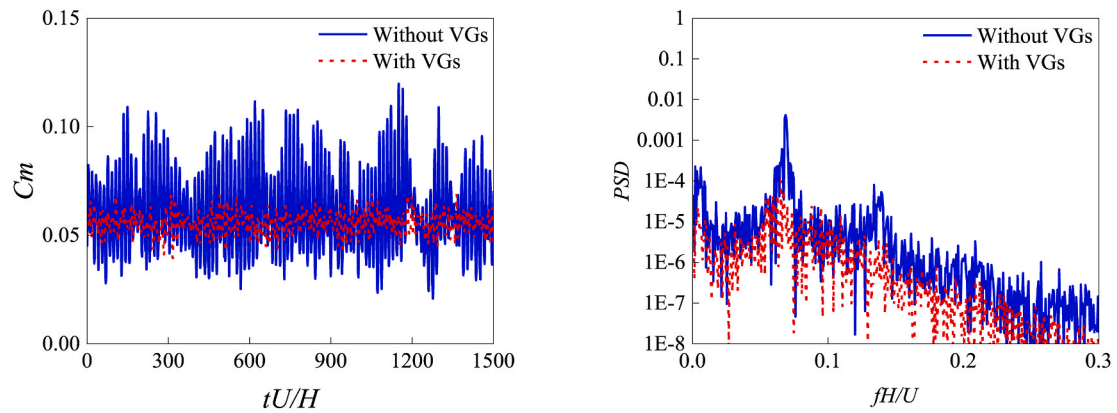


(c) Mesh near the wall



(d) Mesh around VGs

Fig. 22. Description of computational domain.

(a) C_l (b) C_d (c) C_m **Fig. 23.** Time history and PSD of force coefficients.

width of the wall).

Without loss of universality, the terrible AoA of -3° was selected. A boundary layer was built close to the wall to achieve mesh convergence, with the initial y^+ value set to 1. A tetrahedral mesh was applied to refine the rigid body domain, while a triangular prism mesh was used for the outer fluid domain. The maximum grid distance in the spanwise direction was $\Delta z = 0.018D$, which is lower than the minimum spanwise discretization requirement ($\Delta z = 0.1B$) proposed by Tamura et al. (1998). The final mesh count of “without VG” is about 13 million, and that of “VG” is about 28 million.

The Large Eddy Simulation (LES) model is employed in this study. Velocity-inlet and pressure-outlet boundary conditions are applied to the inlet and outlet side, respectively. To maintain consistency with the wind tunnel tests, the uniform inflow velocity U is set to 3 m/s (Reynolds number $Re = UB/\nu = 1.33 \times 10^5$, where ν is the kinematic viscosity and B is the width of the 1:50 sectional model). The turbulence intensity is set to $I = 0.1\%$. The bounded central differencing scheme is used for discretizing the convective terms, and the SIMPLEC algorithm is adopted. Each time step is set to 0.001, with time matching performed using the second-order implicit time integration scheme.

Fig. 22 presents details of the computational domain and mesh configuration.

4.2. Results and discussion

The lift, drag and moment coefficient (C_d , C_l and C_m) are defined as:

$$C_l = \frac{F_l}{0.5\rho U^2 B} \quad (1)$$

$$C_d = \frac{F_d}{0.5\rho U^2 D} \quad (2)$$

$$C_m = \frac{F_m}{0.5\rho U^2 B^2} \quad (3)$$

where ρ is the fluid density, U is the incoming wind speed, D is the height of the wall, B is the width of the wall and F_l is the mean lift force on unit length of the wall, respectively.

To compare the mitigation effect between cases with and without VG, the aerodynamic force data of the two cases were compared and analyzed. Time history and power spectrum density (PSD) of C_l , C_d and C_m are shown in Fig. 23.

As show in Fig. 23, compared with the current girder, the girder with VGs exhibits a lower mean drag force, while the mean lift force and moment force are noticeably higher. Additionally, the aerodynamic force fluctuations in the VG-controlled case are lower than in the baseline case, with all peak values of the PSD being lower than those of the baseline. These observations indicate that the VGs mitigate vortex

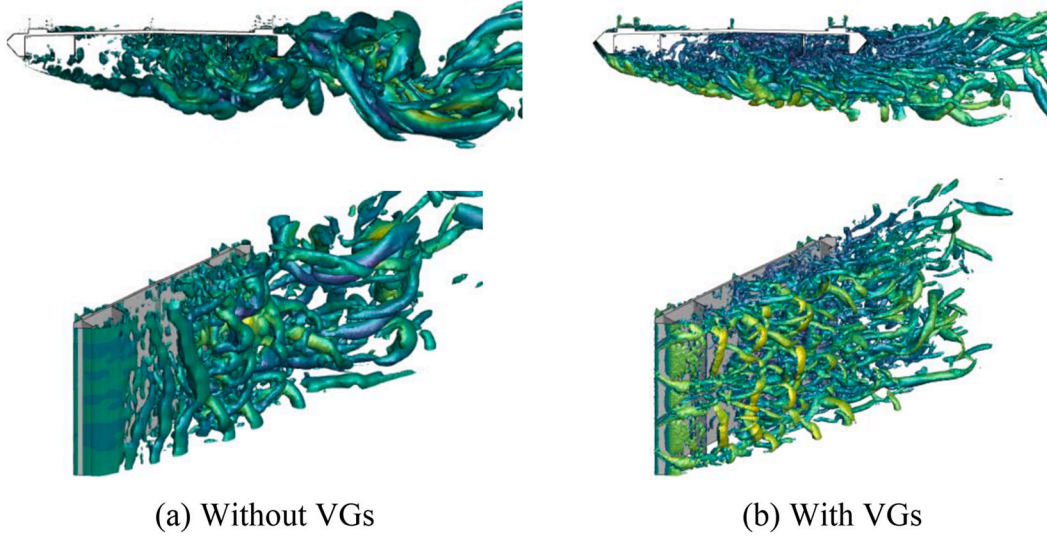


Fig. 24. Instantaneous vortical structures around the girder. $Q = 2000 \text{ s}^{-1}$.

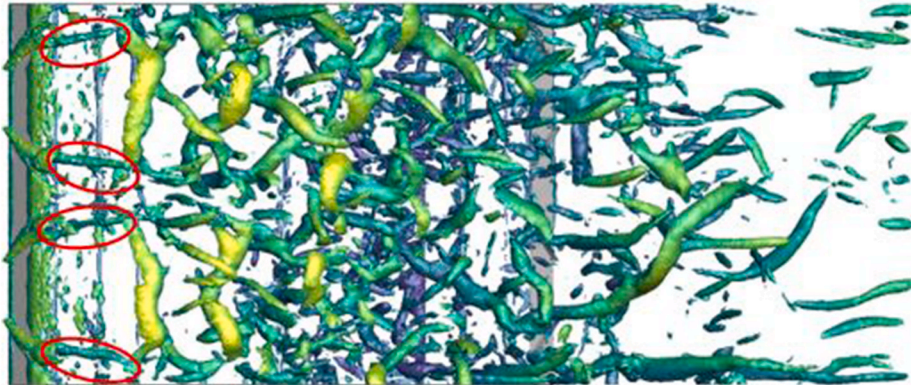


Fig. 25. Streamwise vortical structures (girder with VG). $Q = 4000 \text{ s}^{-1}$.

shedding, creating a condition in which VIV is less likely to occur.

The Q-criterion is applied in this study to identify vortical structures (Hunt, 1988). The identification results for both the current girder and the girder with VGs are shown in Fig. 24: large-scale vortical structures for both cases are located on the lower side and in the near wake of the girder. However, a key difference is observed: the current girder exhibits strong spanwise correlation in its vortical structures, resembling regular Von Kármán vortical structures in the near wake. In contrast, the girder with VGs displays pronounced three-dimensional characteristics, with vortical structures appearing disordered in the near wake. Fig. 25 further highlights that the vortex pairs generated by the VGs create an interaction between streamwise and spanwise vortical structures, disrupting the two-dimensional characteristics of the vortices. This disruption breaks down the spanwise correlations of the vortical structures, enhanced the three-dimensional wake instability and effectively suppressing the VIV.

5. VIV mitigation effect of VG on actual bridge

Based on the results of the wind tunnel tests, it was decided to adopt VGs as the aerodynamic solution to mitigate the unexpected WIVs of the bridge.

The installation process was from September to October 2023. As shown in Fig. 26, the construction team employed a crane-assisted method for partial construction without interrupting the traffic, installing the VGs one by one. Each VG steel plate is supported by three small brackets, directly welded to the lower side of wind fairing. This whole construction process was completed in about one month. The completed VGs are shown in Fig. 27.

Following the installation of the VGs, the bridge experienced multiple instances of winds with wind speeds of approximately 10 m/s; however, according to the data from the health monitoring system,

similar unexpected WIVs did not occur again. Even after a year of continuous monitoring, the VG still demonstrate excellent WIV mitigation effects.

6. Conclusions

Unexpected wind-induced vibrations (WIV) with considerable amplitude of the first vertical mode were observed on a cable-stayed bridge with a II-shaped steel-concrete composite girder after nine years of service. Based on on-site inspection, field measurements, wind tunnel tests and CFD simulations, the possible causes were investigated, and aerodynamic solution based on vortex generators (VG) was proposed and practiced. Main conclusions are summarized as follows.

- (1) The bridge damping obtained through field measurement was as low as 0.2–0.4%, differing significantly from the assumed values of the recognized design codes, which was also confirmed as the main cause of the observed unexpected wind-induced vibration of the bridge. Where there is accurate knowledge of the structural damping, it is possible to obtain good agreement of VIV response between wind tunnel test and on-site measurement. The main difficulty is that the damping is often not known with any precision in the design stage, and have potential possibility of change during the long service period. Field measurement of damping after construction can be helpful in this regard. Also, long-term monitoring of damping would be highly beneficial.
- (2) The changing of central guardrail has no evident impact on the VIV performance of the bridge girder since the guardrail kept very high porosity.
- (3) The vortex shedding caused VIV mainly exists at the lower edge and in the near wake of the girder, and their spanwise correlation can be disrupted by the streamwise vortices generated by the VGs



Fig. 26. Construction of VG

Fig. 27. VG installed on the actual bridge.

installed on the lower sides of wind fairings, thereby mitigating the VIV. In the present engineering practice, the effectiveness of the VGs in mitigating VIV was verified by long-term monitoring of the bridge.

CRedit authorship contribution statement

Chaoqun Wang: Writing – review & editing, Methodology, Investigation, Data curation. **Haoyu Xu:** Writing – original draft, Visualization, Software. **Zhiwen Huang:** Project administration, Methodology. **Zhouquan Feng:** Formal analysis, Data curation. **Qi Li:** Resources, Project administration. **Xugang Hua:** Writing – review & editing, Supervision, Methodology, Funding acquisition, Conceptualization. **Zhengqing Chen:** Supervision, Conceptualization.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Data availability

Data will be made available on request.

References

- Bai, H., Ji, N., Xu, G., Li, J., 2020. An alternative aerodynamic mitigation measure for improving bridge flutter and vortex induced vibration (VIV) stability: sealed traffic barrier. *J. Wind Eng. Ind. Aerod.* 206.
- Barni, N., Mannini, C., 2024. Parametric effects of turbulence on the flutter stability of suspension bridges. *J. Wind Eng. Ind. Aerod.* 245.
- Battista, R.C., Pfeil, M.S., 2000. Reduction of vortex-induced oscillations of Rio-Niteroi bridge by dynamic control devices. *J. Wind Eng. Ind. Aerod.* 84, 273–288.
- Cao, Y., Huang, Z., Zhang, H., Hua, X., Chen, Z., Wan, T., 2023. Discrete viscous dampers for multi-mode vortex-induced vibration control of long-span suspension bridges. *J. Wind Eng. Ind. Aerod.* 243.
- Choi, H., Jeon, W.-P., Kim, J., 2008. Control of flow over a bluff body. *Annu. Rev. Fluid Mech.* 40, 113–139.
- Corriols, A.S., Morgenthal, G., 2012. Computational analysis of VIV observed on existing bridges. *The Seventh International Colloquium on Bluff Body Aerodynamics and Applications (BBAA7)*, pp. 2–6. Shanghai, China.
- Diana, G., Fiammenghi, G., Belloli, M., Rocchi, D., 2013. Wind tunnel tests and numerical approach for long span bridges: the Messina bridge. *J. Wind Eng. Ind. Aerod.* 122, 38–49.
- Dong, J., Huang, L., Liao, H., Wang, Q., 2021. Investigation on suppressing vortex-induced vibrations of the rectangular steel box girder for railway cable-stayed bridges by installing wind fairings. *J. Wind Eng. Ind. Aerod.* 219.
- Doddipatla, L.S., Hangan, H., Durgesh, V., Naughton, J., 2017. Wake dynamics resulting from trailing-edge spanwise sinusoidal perturbation. *AIAA J.* 55, 1833–1851.
- El-Gammal, M., Hangan, H., King, P., 2007. Control of vortex shedding-induced effects in a sectional bridge model by spanwise perturbation method. *J. Wind Eng. Ind. Aerod.* 95, 663–678.
- Elias, S., Matsagar, V., 2017. Research developments in vibration control of structures using passive tuned mass dampers. *Annu. Rev. Control* 44, 129–156.
- European Commission, 2005. BS EN 1991-1-4: 2005, Eurocode 1-Actions on Structures-Part 1-4: General Action-Wind Actions.
- Frandsen, J.B., 2001. Simultaneous pressures and accelerations measured full-scale on the Great Belt East suspension bridge. *J. Wind Eng. Ind. Aerod.* 89, 95–129.
- Fujino, Y., Siringoringo, D., 2013. Vibration mechanisms and controls of long-span bridges: a review. *Struct. Eng. Int.* 23, 248–268.
- Fujino, Y., Siringoringo, D.M., Ikeda, Y., Nagayama, T., Mizutani, T., 2019. Research and implementations of structural monitoring for bridges and buildings in Japan. *Engineering* 5, 1093–1119.
- Fujino, Y., Yoshida, Y., 2002. Wind-induced vibration and control of trans-tokyo Bay crossing bridge. *J. Struct. Eng. Asce* 128, 1012–1025.
- Gao, D., Deng, Z., Yang, W., Chen, W., 2021. Review of the excitation mechanism and aerodynamic flow control of vortex-induced vibration of the main girder for long-span bridges: a vortex-dynamics approach. *J. Fluid Struct.* 105.
- Ge, Y., Zhao, L., Cao, J., 2022. Case study of vortex-induced vibration and mitigation mechanism for a long-span suspension bridge. *J. Wind Eng. Ind. Aerod.* 220.
- Hua, X., Huang, Z., Chen, Z., 2019. Multi-mode vertical vortex-induced vibration of suspension bridges and control strategy. *China J. Highw. Transp.* 32, 1001–7372 (in Chinese).
- Hua, X., Wang, C., Li, S., Chen, Z., 2020. Experimental investigation of wind-induced vibrations of main cables for suspension bridges in construction phases. *J. Fluid Struct.* 93.
- Hwang, Y.C., Kim, S., Kim, H.-K., 2020. Cause investigation of high-mode vortex-induced vibration in a long-span suspension bridge. *Struct. Infrastruct. Eng.* 16, 84–93.
- Hui, Y., Tang, Y., Yang, Q., Chen, B., 2024. A Wake-Oscillator Model for predicting VIV of 4-to-1 rectangular section cylinder. *Nonlinear Dyn.* 112, 8985–8999.
- Hunt, J., 1988. Eddies Stream, and Convergence Zones in Turbulent Flows. Center for Turbulence Research Report CTR-s88, pp. 193–208.
- Han, Y., Mi, L.H., Shen, L., Cai, C.S., Liu, Y.C., Li, K., Xu, G.J., 2022. A short-term wind speed prediction method utilizing novel hybrid deep learning algorithms to correct numerical weather forecasting. *Appl. Energy* 312, 118777.
- Larsen, A., Esdahl, S., Andersen, J.E., Vejrum, T., 2000. Storebælt suspension bridge -: vortex shedding excitation and mitigation by guide vanes. *J. Wind Eng. Ind. Aerod.* 88, 283–296.
- Larsen, A., Larose, G.L., 2015. Dynamic wind effects on suspension and cable-stayed bridges. *J. Sound Vib.* 334, 2–28.
- Li, H., Laima, S., Jing, H., 2014a. Reynolds number effects on aerodynamic characteristics and vortex-induced vibration of a twin-box girder. *J. Fluid Struct.* 50, 358–375.
- Li, H., Laima, S., Zhang, Q., Li, N., Liu, Z., 2014b. Field monitoring and validation of vortex-induced vibrations of a long-span suspension bridge. *J. Wind Eng. Ind. Aerod.* 124, 54–67.
- Li, K., Qian, G., Ge, Y., Zhao, L., Di, J., 2019. Control effect and mechanism investigation on the horizontal flow-isolating plate for PI shaped bridge decks' VIV stability. *Wind Struct.* 28, 99–110 (in Chinese).
- Li, M., Sun, Y., Jing, H., Li, M., 2018. Vortex-induced vibration optimization of a wide streamline box girder by wind tunnel test. *KSCE J. Civ. Eng.* 22, 5143–5153.
- Li, M., Yu, J., Lin, Z., Li, M., 2024. Theoretical and experimental study of the spanwise effect of turbulence on the aerodynamic lift on a wing with different aspect ratios. *Phys. Fluids* 36.
- Li, T., Liang, H., Zhang, J., Zhang, J., 2023. Numerical study on aerodynamic resistance reduction of high-speed train using vortex generator. *Eng. Appl. Comput. Fluid Mech.* 17.
- Lin, J.C., 2002. Review of research on low-profile vortex generators to control boundary-layer separation. *Prog. Aero. Sci.* 38, 389–420.
- Liu, Z., He, F., Yan, A., Liu, Z., Yin, T., Chen, Z., 2023a. Mechanism and aerodynamic countermeasures of vortex-induced vibration of a cable-stayed bridge with narrow T-shaped girder sections. *J. Bridge Eng.* 28.
- Liu, L., Zou, Y., He, X., 2023b. Effects of wind barriers on VIV performances of twin separated parallel decks for a long-span rail-cum-road bridge. *J. Wind Eng. Ind. Aerod.* 236, 105367.
- Ma, C., Pei, C., Liao, H.-L., Liu, M., Li, M., 2020. Field measurement and wind tunnel study of aerodynamic characteristics of twin-box girder. *J. Wind Eng. Ind. Aerod.* 202.
- MacDonald, J.H.G., Irwin, P.A., Fletcher, M.S., 2002. Vortex-induced vibrations of the Second Severn Crossing cable-stayed bridge - full-scale and wind tunnel measurements. *Proc. Institution of Civil Engineers-Struct. Buildings* 152, 123–134.
- Magalhaes, F., Caetano, E., Cunha, A., 2007. Challenges in the application of stochastic modal identification methods to a cable-stayed bridge. *J. Bridge Eng.* 12, 746–754.
- Meng, H., Chen, G.B., Gao, D.L., 2024. Aerodynamics and surrounding flow patterns of a long-span bridge girder model with triple-separated boxes. *Phys. Fluids* 36, 035134.
- Meng, H., Chen, W.L., Li, H., 2023. The vortex-induced vibrations of a triple-box girder model of a long-span rail-cum-road bridge. *J. Wind Eng. Ind. Aerod.* 235, 105359.
- Matsumoto, M., Shiraiishi, N., Shirato, H., Stoyanoff, S., Yagi, T., 1993. Mechanism of, and turbulence effect on vortex-induced oscillations for bridge box girders. *J. Wind Eng. Ind. Aerod.* 49, 467–476.
- Nagao, F., Utsunomiya, H., Yoshioka, E., Ikeuchi, A., Kobayashi, H., 1997. Effects of handrails on separated shear flow and vortex-induced oscillation. *J. Wind Eng. Ind. Aerod.* 69–71, 819–827.
- Raghavan, K., Bernitsas, M.M., 2011. Experimental investigation of Reynolds number effect on vortex induced vibration of rigid circular cylinder on elastic supports. *Ocean Eng.* 38, 719–731.
- Ran, Y., Chen, W.L., Jin, Y., Li, H., Gao, D.L., 2024. Reynolds number sensitivity of the vortex dynamics around a long-span rail-cum-road bridge girder with three separated boxes. *J. Wind Eng. Ind. Aerod.* 250, 105785.
- Schewe, G., Larsen, A., 1998. Reynolds number effects in the flow around a bluff bridge deck cross section. *J. Wind Eng. Ind. Aerod.* 74–6, 829–838.
- Tamura, T., Miyagi, T., Kitagishi, T., 1998. Numerical prediction of unsteady pressures on a square cylinder with various corner shapes. *J. Wind Eng. Ind. Aerod.* 74–6, 531–542.

- Tang, Y., Hua, X.G., Chen, Z.Q., Zhou, Y., 2019. Experimental investigation of flutter characteristics of shallow T-section at post-critical regime. *J. Fluid Struct.* 88, 275–291.
- Tao, S.B., Tang, A.P., Xin, D.B., Liu, K.T., Zhang, H.F., 2016. Vortex-induced vibration suppression of a circular cylinder with vortex generators. *Shock Vib.* 2016, 1–10.
- The People's Ministry of Transport of China, 2019. Wind-resistant Design Specification for Highway Bridges (JTG/T 3360-01-2018).
- Van Hooff, T., Blocken, B., Tominaga, Y., 2017. On the accuracy of CFD simulations of cross-ventilation flows for a generic isolated building: comparison of RANS, LES and experiments. *Build. Environ.* 114, 148–165.
- Viviescas Jaimes, A., Carrillo León, W.J., Vargas Carvajal, L.A., 2018. Structural health monitoring baseline of Gómez Ortiz bridge using ambient vibration tests. *Inge Cuc* 14, 52–65.
- Wang, C., Hua, X., Feng, Z., Xiao, H., Chen, Z., Zhang, J., 2021. Experimental investigation on vortex-induced vibrations of a triple-box girder with web modification. *J. Wind Eng. Ind. Aerod.* 218.
- Wang, C., Wen, Q., Zhou, S., Hua, X., Huang, Z., Chen, Z., 2022a. Effects of end condition and aspect ratio on vortex-induced vibration of a 5:1 rectangular cylinder. *J. Fluid Struct.* 109.
- Wang, C., Huang, Z., Hua, X., He, X., Zhou, S., Chen, Z., 2022b. Aerodynamic mechanism of triggering and suppression of vortex-induced vibrations for a triple-box girder. *J. Wind Eng. Ind. Aerod.* 227.
- Williamson, C.H.K., 1996a. Three-dimensional wake transition. *J. Fluid Mech.* 328, 345–407.
- Williamson, C.H.K., 1996b. Vortex dynamics in the cylinder wake. *Annu. Rev. Fluid Mech.* 28, 477–539.
- Williamson, C.H.K., Raghuraman, N.G., 2004. Vortex-induced vibrations. *Annu. Rev. Fluid Mech.* 36, 413–455.
- Xie, Z.Y., He, X.H., Wang, H.F., Huang, Z.R., Li, H., 2024. Aerodynamic characteristics of a high-speed train on flat-box girder for different yaw angles. *J. Wind Eng. Ind. Aerod.* 250.
- Xin, D., Zhang, H., Ou, J., 2018. Experimental study on mitigating vortex-induced vibration of a bridge by using passive vortex generators. *J. Wind Eng. Ind. Aerod.* 175, 100–110.
- Xin, D., Zhan, J., Zhang, H., Ou, J., 2021. Control of vortex-induced vibration of a long-span bridge by inclined railings. *J. Bridge Eng.* 26.
- Yan, X., Ye, J., Zhao, Z., Hu, X., Liao, L., 2011. Effect of vortex generators on aerodynamic characteristics of a car. 2nd International Conference on Advances in Materials and Manufacturing Processes (ICAMMP 2011). PEOPLES R CHINA, Guilin, p. 1873.
- Yang, Y., Kim, S., Hwang, Y., Kim, H.-K., 2021. Experimental study on suppression of vortex-induced vibration of bridge deck using vertical stabilizer plates. *J. Wind Eng. Ind. Aerod.* 210.
- Yu, E.B., Xu, G.J., Han, Y., Hu, P., Townsend, J.F., Li, Y.L., 2022. Bridge vibration under complex wind field and corresponding measurements: a review. *J. Traffic and Transport. Eng. English Edition* 9, 339–362.
- Zhang, H., Xin, D., Ou, J., 2019. Wake control using spanwise-varying vortex generators on bridge decks: a computational study. *J. Wind Eng. Ind. Aerod.* 184, 185–197.
- Zhang, H., Zhang, H., Xin, D., Zhan, J., Wang, R., Zhou, L., 2022. Vortex-induced vibration control of a streamline box girder using the wake perturbation of horizontal axis micro-wind turbines. *J. Fluid Struct.* 108.
- Zhan, J., Xin, D., Ou, J., Liu, Z., 2020. Experimental study on suppressing vortex-induced vibration of a long-span bridge by installing the wavy railings. *J. Wind Eng. Ind. Aerod.* 202.
- Zhao, L., Cui, W., Shen, X., Xu, S., Ding, Y., Ge, Y., 2022. A fast on-site measure-analyze-suppress response to control vortex-induced-vibration of a long-span bridge. *Structures* 35, 192–201.
- Zhu, S.Y., Yi, R., Li, Y.L., Xu, X.Y., 2024. Buffeting reliability of high-rise bridge tower in mountain area based on CNN-BiLSTM. *Appl. Soft Comput.* 163, 111872.